



CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

End Semester Examination Dec-2025

Programme: M.Sc (AI)

Course Title: Indian Knowledge System in Computer Science

Maximum points: 70

Enrollment No. : CUSB2502332023

Course code: CAI8IDC00402

Duration: 2:00 Hrs

Semester - 1st

Instructions:

1. Attempt questions in all the Sections as directed.

Section A

*Answer all the questions in a maximum of 20 words, each carrying 2 Points.
(5 X 2 = 10)*

1. Which dynasty pioneered the preparation of wootz steel in India?
2. As per the current knowledge whose text mentions fibonacci series?
3. Which are the top three Indian IT companies in terms of market capital?
4. Who was the head of the committee that recommended establishing IITs in India and when this committee was formed?
5. List three objectives of establishing AICTE.

Section B

*Answer any 4 questions below in a maximum of 200 words, each carrying 10 points.
(4 X 10 = 40)*

1. Describe the contribution of Aryabhatta to Indian mathematics and astronomy.
2. Discuss all the evidence of zero found in India.
3. Write a brief note on the following two ancient Indians who significantly contributed to Computing.
[5+5]
 - i. Pingala and his contribution
 - ii. Brahmagupta and his contribution to mathematics
4. Explain R&D ecosystem of Computer Science and Information Technology in India.
5. Discuss about the ecosystem of funding for the development of science and technology in India.

Section C

Answer any 1 question below in a maximum of 600 words, each carrying 20 points.

(1 X 20 = 20)

1. Answer the following each carrying 10 points:

- a. Explain why the evolution of counting was necessary to the evolution of human civilizations.
- b. Explain the presence of Fibonacci numbers in the family tree of Honeybees.

2. Answer the following each carrying 10 points:

- a. Discuss how India became a global leader in Information Technology.
- b. Elaborate about the academic ecosystem of Computer Science and Information Technology in India.